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Spatially resolved modeling of pumped storage and hydropower for China's carbon neutrality

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China's commitment to carbon neutrality by 2060 requires 5000–8000 GW integration of variable renewable energy, which may pose unprecedented grid stability challenges. Using spatial-temporally resolved modeling to co-optimize capacity expansion and system operation for hybrid wind–solar–hydropower systems, we quantify the flexibility requirements of pumped hydro storage (PHS) and cascaded hydropower in a decarbonized grid. We demonstrate that deploying 205 GW of open-loop PHS reduces system costs by 81 billion yuan (about 11.6 billion USD) annually while decreasing VRE curtailment by 11%. Through reservoir-level modeling of closed-loop PHS, we reveal that conventional approaches systematically overestimate capacity requirements by 30% and inflate costs by 36 billion yuan (about 5.2 billion USD) annually by failing to capture operational flexibility. Crucially, price volatility in renewable-dominated markets generates arbitrage revenues sufficient to enable the commercial viability of PHS without subsidies—a finding with global implications for financing energy transitions. Optimizing hydropower cascade operations yields an additional 116 billion yuan (about 16.6 billion USD) in annual savings. Multi-year climate simulations confirm system resilience across diverse weather conditions. Sophisticated modeling of pumped storage and hydropower flexibility is essential for designing economically viable pathways to deep decarbonization in large-scale power systems.

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Broader context

Activating the latent flexibility within hydrological systems could accelerate the transition to a reliable net-zero power grid, yet current power models often decouple water resource management from electricity market dynamics. This study bridges that gap by integrating reservoir-level cascade dynamics and site-specific pumped storage optimization into a continental-scale power system model. We demonstrate that coordinating open-loop pumped storage with hydropower cascades could enhance system flexibility. Our analysis also reveals a pivotal economic insight: the intrinsic price volatility of renewable-dominated grids creates sufficient arbitrage revenue to render pumped storage commercially viable without government subsidies. This work offers a rigorous roadmap for policymakers to leverage hydrological infrastructure for deep decarbonization, validating a pathway from subsidy-dependent planning to market-driven grid stability.

1 Introduction

In response to global climate change and sustainable development challenges, many countries and regions have established targets for renewable energy installations, particularly wind and solar power, to phase down fossil fuels.^{1–6} China, as the world's largest emitter of electricity-related greenhouse gases,⁷ announced its ambitious plan in 2020 to achieve carbon

neutrality by 2060.⁸ In pursuit of this goal, the nation has prioritized the development of renewable energy. China achieved its target of 1200 GW of installed wind and solar capacity in 2024, six years ahead of the original 2030 schedule,^{9,10} and then updated the target to 3600 GW before 2035 for its nationally determined contributions in 2025.¹¹ However, previous research indicates that integrating multi-terawatt-scale variable renewable energy (VRE) into power grids poses significant challenges to system stability,⁴ e.g., possible supply shortages during peak-demand periods. Among potential solutions to this challenge, pumped hydro storage (PHS) and hydropower have emerged as promising regulatory resources for their ability to regulate power supply temporally through reservoir management while maintaining zero carbon attributes.^{12–15} Meanwhile, China possesses the world's largest

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potential of pumped storage and hydropower capacity,^{16–18} which will be of great significance to the nation's transition to a clean electricity system. Therefore, systematically understanding the roles of pumped storage and hydropower in China's VRE-dominated power system is critical for advancing long-term planning and policy development.

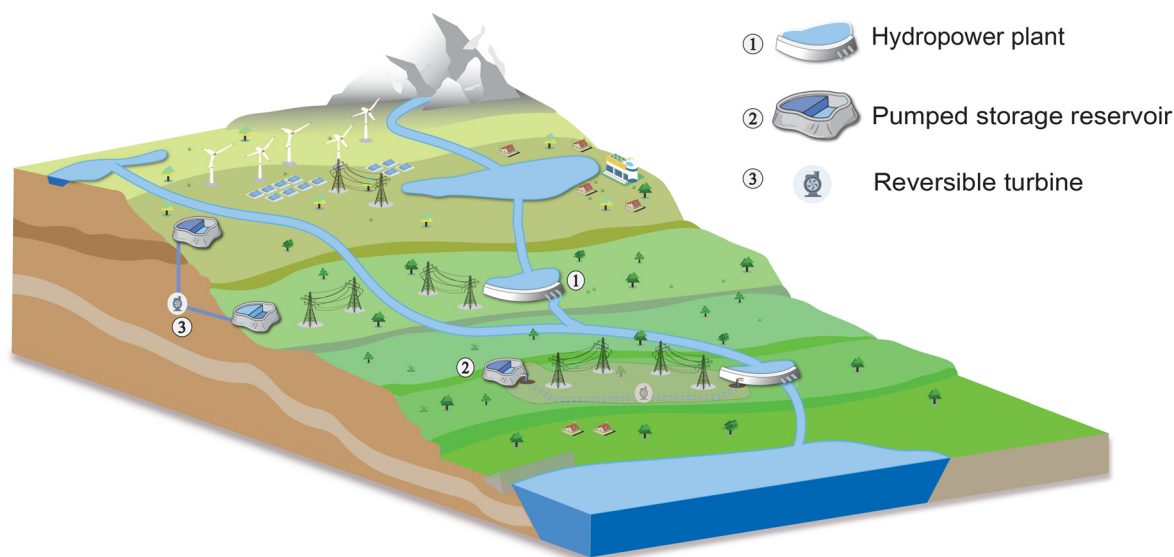
Power system expansion models (PSEMs), generally formulated as linear programming problems to minimize annualized system costs while satisfying operational constraints and decarbonization policies,¹⁹ are widely used analytical tools in research on power system transitions.^{1–3,5,20,21} Developing PSEMs that can systematically analyze the expansion and operational mechanisms of pumped storage and hydropower with simultaneous large-scale deployment of weather-dependent resources for a large region like China entails several critical challenges. First, nationwide geographical coverage with cross-region power balance optimization is essential¹ because renewable resources and electricity demand are overall spatially mismatched in China, exemplified by abundant wind and solar resources in Northwest China, while the electricity demand concentrates along the eastern coast.²² Second, fine temporal resolution (≤ 1 h) and extended temporal coverage (8760 h) are important for capturing the variability of renewable energies and optimizing the operation of multiple generation technologies to more accurately represent the complementarities between hydroelectricity (including pumped storage) and VREs. Third, high spatial resolution resource assessment and siting optimization of renewable energies are imperative to adequately represent hydropower cascade operations, coordinate charge/discharge routes for pumped storage at the reservoir level, and address the spatial heterogeneity inherent in wind and solar outputs. Fourth, an integrated optimization among renewable energies, transmission, energy storage, and conventional power (e.g., coal, natural gas, nuclear, and biomass energy) for analyzing their synergistic coordination is much needed. These challenges stem from both data integration complexities and computational demands of high-resolution modeling across spatial, temporal, and technological dimensions.

Existing long-term planning models designed to analyze the transition of China's power system and the role of the hybrid wind-solar-hydropower system have limitations in adequately addressing these challenges. First, numerous PSEMs adopt a two-stage optimization approach: initially simulating power system operation using representative hours (generally dozens to hundreds) for capacity expansion and then fixing the capacity to optimize power operation over longer temporal coverage (e.g., 8760 h).^{2,20,22,23} This stepwise optimization approach has been proven to introduce non-negligible inaccuracies,^{4,6,19} as the capacity expansion is only informed by the selected representative periods, which may lead to limited representation of the intra-annual variability in power supply and demand and thus potentially result in suboptimal investment planning.^{24,25} Second, the optimization resolution for wind, solar, and hydropower is often overly simplified. For wind and solar photovoltaics (PV), though the spatial heterogeneity and grid integration costs are critical factors that affect planning decisions,^{3,4,6} previous models usually aggregate VRE cells in a provincial

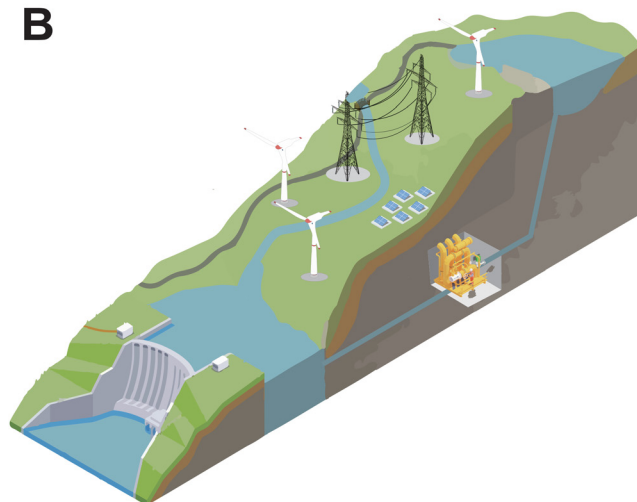
grid to several resource regions^{1,2,20,22,26,27} to reduce computational complexity, while may introduce bias in optimal capacity deployment.^{3,4,6} Similarly, many PSEMs represent the hydropower with flexible reservoir capacity that can provide stable grid regulation only at the resolution of the provincial level using a layer dispatch method, dismissing its valuable dispatchability.^{1,3,20,21,28} Additionally, real-world cases indicate that hierarchical co-optimization of hydropower cascades could enable efficient balancing of renewable energy supply and grid demand, which cannot be represented in a provincial layer model.^{29,30} Although some recent studies (e.g., ref. 2 and 12) optimized the reservoir management of hydropower at the dam-site level (only including the south grid in ref. 12), their models only account for existing installed hydropower without new capacity expansion and adopt representative hours to simulate a whole-year operation. Third, in previous optimization models, pumped storage is often simplified with modeling approaches similar to chemical energy storage—planning capacity and charge/discharge routes at the provincial level, differentiating from other storage technologies merely through investment cost and charge/discharge efficiency parameters.^{1,2,20,21,23,27,28} Besides, open-loop pumped storage that can be integrated with hydropower (sharing one reservoir) to enhance hydropower's flexibility has not yet been modeled in previous works. Therefore, a refined integrated framework is essential to optimize the hybrid systems combining hydropower, wind, solar, and pumped storage, addressing critical modeling gaps and investigating their role in decarbonizing China's power system toward carbon neutrality.

In this study, we develop a temporally and spatially resolved PSEM with hourly resolution over a full planning year to explore the role of hybrid wind, solar, hydropower, and pumped storage (including open-loop and closed-loop, see Fig. 1) in China's negative emission power systems around 2060. We extend our China Integrated Sustainable Power-system Optimization (CISPO) model by introducing a new geographical data integration and modeling approach related to hydropower and pumped storage and maintain the high-resolution representation of other technologies compared to its previous version (e.g., grid cell level expansion and operation for VREs).⁴ First, the CISPO here provides a refined evaluation of China's reservoir hydropower capacity potential (approximately 715 GW) and implements cascade-based capacity expansion and operational optimization for medium to large hydropower plants exceeding 50 MW, a significant improvement over the previous version, which only performed non-cascade dam-site level optimization. This refined operational framework encompasses not only hourly generation scheduling but also rigorous reservoir water level management, environmental flow constraints, and ancillary service provision for reserve capacity. Second, leveraging high spatial resolution reservoir data,¹⁸ the CISPO here assessed potential open-loop pumped storage that can co-utilize reservoirs with existing hydropower facilities, incorporating these into cascade optimization scenarios to extend hydropower applications beyond what was previously modeled. Third, based on site-specific resource assessments and cost estimates,

A



B



C

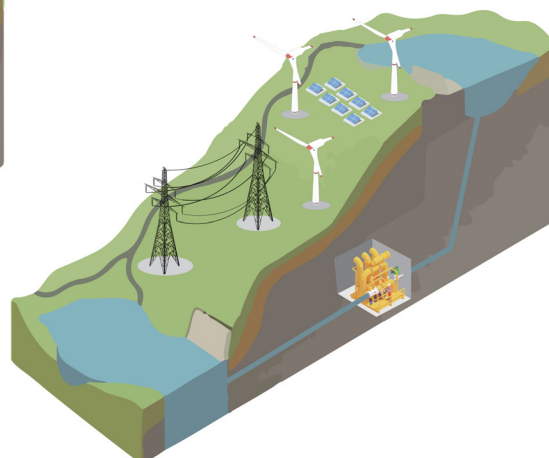


Fig. 1 Optimization schema of pumped storage and hydropower in this study. (A) Hybrid wind-solar-hydropower system (including pumped storage). (B) Open-loop pumped storage system that shares one reservoir (upper or lower) with hydropower. (C) Closed-loop pumped storage system.

this study employs spatially differentiated investment cost data for both hydropower and pumped storage to reflect regional construction cost variations, replacing the uniform national cost assumptions used in other studies.^{2–4,20,22,27} Fourth, the CISPO here co-optimizes the capacity expansion and charge/discharge route for closed-loop pumped storage at the reservoir site and hourly level, rather than the provincial grid level in the previous version.

With this temporally and spatially enhanced CISPO model, our research addresses several interrelated questions significant to the transition of China's power system toward negative emissions: What role will open- and closed-loop pumped storage play in the hybrid wind-solar-hydro system for China's power sector decarbonization? Is additional financial support needed for pumped storage facilities to strengthen power system regulation? How can cascade-based hydropower

optimization reduce the system costs? By integrating detailed hydropower and pumped storage modeling, the scale of the CISPO model developed in this study reaches approximately 1.3×10^8 constraints, 1.0×10^8 variables, and 6.8×10^8 non-zero parameters, thereby enabling unprecedented operational fidelity in modeling complex hydro-renewable synergies across China's heterogeneous landscapes.

2 Results

2.1 Resource potential and scenario design

As the key input resource, we adopt a geographical information system with high-resolution digital elevation model datasets at 90 m,³¹ a natural discharge dataset for global rivers spanning from 1980 to 2020³² and a reservoir potential dataset¹⁸ in China

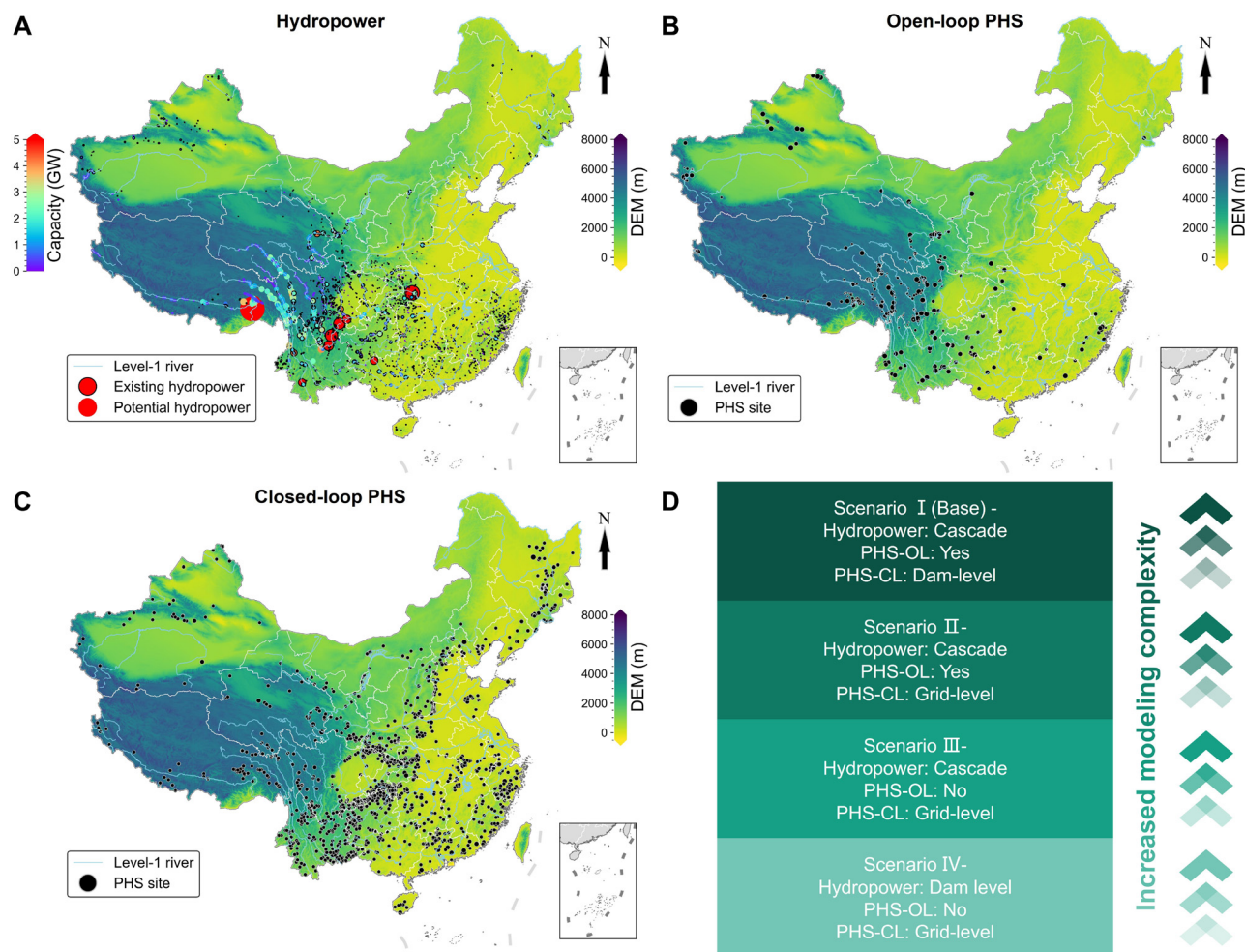


Fig. 2 Installation potential of hydropower and pumped storage used as input and scenario design in this study. (A) Existing installed and potential hydropower plants. (B) Installation potential of open-loop pumped hydro storage (PHS: OL). (C) Installation potential of closed-loop pumped hydro storage (PHS: CL). (D) Scenario matrix (PHS-OL: open-loop pumped hydro storage; PHS-CL: closed-loop pumped hydro storage).

to assess the technical installation potential of hydropower and pumped storage, with the results shown in Fig. 2. Our analysis excludes environmentally sensitive areas, such as nature reserves, and incorporates the economic costs of population displacement according to the hydropower station's inundation area.¹⁷ The results indicate that China's established and potential hydropower capacity totals 715 GW, predominantly concentrated in the southwestern regions, including Xizang (222 GW), Sichuan (186 GW), Yunnan (85 GW), Qinghai (50 GW), and Hubei (34 GW). See SI Table S1 and Fig. S2 for the region division used in this study. These provinces collectively account for 577 GW of hydropower capacity, representing over 80% of the national potential. Assuming that hydropower stations with installed capacities exceeding 50 MW (the threshold for small-sized hydropower according to ref. 33) can be connected to another reservoir for storage enhancement, we identify a nationwide potential of 390 GW (270 reservoirs) for the open-loop pumped storage systems. This potential is primarily distributed across Xizang (98 GW), Sichuan (77 GW), and Yunnan (55 GW), showing a strong correlation with reservoir

hydropower potential. Following economic, ecological, and environmental screening criteria,¹⁷ the national installation potential for closed-loop pumped storage systems is approximately 1960 GW, concentrated in provinces conducive to creating reservoirs with substantial elevation differences, notably Yunnan (298 GW), Guizhou (200 GW), Sichuan (168 GW), and Chongqing (130 GW). We show a detailed description of this resource assessment in the methods and the SI Sections S2 and S3.

In this study, we design four scenarios to test the impacts of the modeling framework on China's power system analysis, as shown in Fig. 2D. Four scenarios related to hydropower and pumped storage modeling are classified according to the model scale. The base scenario (scenario I) not only encompasses all features from the previous version of CISPO⁴ but also incorporates several significant enhancements. First, it implements cascaded capacity expansion and operational scheduling for medium-to-large hydropower plants exceeding 50 MW³³ (691 GW in total, covering over 96% hydropower capacity). Second, the model considers reservoir-level open-loop pumped

hydro storage expansion, which forms a hybrid system with conventional hydropower. Third, it elevates the optimization of closed-loop pumped hydro capacity expansion and dispatch to the reservoir-pair level. To the best of our knowledge, this detailed representation of hydropower and pumped hydro storage achieves the highest fidelity to date. Building upon this framework, we systematically change specific modeling details to analyze their impact on the results. These alternative cases include Scenario II, which optimizes closed-loop pumped storage operations at the grid level; Scenario III, which prohibits the construction of open-loop pumped storage; and Scenario IV, which optimizes hydropower capacity and operation solely at the individual dam level. This scenario effectively simulates a “fragmented governance” situation where coordination between upstream and downstream operators is inhibited by institutional barriers, contrasting with the “integrated management” assumption in the base scenario.

2.2 Hybrid wind-solar-hydropower system in 2060

Fig. 3 presents the optimization results for China's integrated wind-solar-hydro system in the base scenario for 2060, encompassing the optimal capacity allocations and generation profiles for hydropower and pumped storage facilities (both open-loop and closed-loop configurations), grid-cell-level spatial distributions of wind and solar PV installations, system cost of electricity (SCOE, excluding transmission, distribution, and administrative expenses), and generation-capacity mix configurations at both provincial and national scales. Our analysis demonstrates that large-scale integration of wind, solar, hydropower, and biomass energy can meet the majority of China's projected annual electricity demand of 20 PWh while simultaneously achieving negative emissions targets (-550 MtCO_2 per year^{4,34}) by 2060. The optimized hybrid system configuration comprises 690 GW of hydropower capacity (generating 3.19 PWh per year), 2727 GW of wind power (7.8 PWh per year), 5178 GW of solar PV (7.1 PWh per year), and 181 GW of biomass energy (0.84 PWh per year). These installed capacities yield a total annual national generation exceeding 18 PWh, including curtailed wind and solar output. Provincial-level annual generation profiles are provided in the SI, Fig. S20, while national-scale hourly supply-demand dynamics are illustrated in the SI, Fig. S21. To facilitate the integration of these VRE sources, the optimization model prescribes 368 GW of closed-loop pumped storage capacity and 205 GW of open-loop pumped storage capacity, supplemented by an additional 626 GW of battery storage (BAT) deployed strategically in regions with limited pumped storage potential. The system achieves remarkably low curtailment rates nationwide—5.4% for wind and 6.1% for solar—reflecting efficient VRE integration. Leveraging China's geographically diverse renewable resource endowment and robust transmission infrastructure, this configuration maintains strong economic viability, with an SCOE of approximately 0.33 yuan per kWh.

China's hybrid wind-solar-hydropower systems exhibit distinct spatial distribution patterns, as illustrated in Fig. 3A–D. Onshore wind power is concentrated in the Three-North regions (North, Northeast, and Northwest China), while

offshore wind power clusters along the southeastern coastal areas. Solar installations are distributed widely in densely populated eastern and southern regions, with additional concentrated large-scale bases emerging in the northwest. Hydropower generation is concentrated primarily in Southwest and Central China, constituting the dominant power source for the future west-to-east electricity transmission. In the base scenario with national cascade optimization of hydropower systems, we find that the required 690 GW of hydropower capacity could generate approximately 3.19 PWh of clean electricity annually. This capacity represents approximately double the current installation, with new hydropower stations predominantly comprising large-scale facilities in regions with superior water resources. Therefore, the national average annual capacity factors slightly exceed historical values (equivalent generation hours of 4300 hours, compared to 3726 hours in 2019). According to China's nine primary river basin classifications, the Yangtze River Basin demonstrates the largest installed capacity at 326 GW (114 GW new), followed by the Southwest Basin at 244 GW (216 GW new). Provincial results reveal that Xizang would deploy 205 GW of hydropower, positioning it as the nation's largest hydropower province with an annual generation of 904 TWh, complementing its 180 GW solar installations as primary sources for electricity export, see electricity flow alongside transmission lines in Fig. 4A. Sichuan would deploy 184 GW of hydropower (approximately 898 TWh per year), serving not only as a major source for China's west-to-east transmission but also as a crucial clean electricity supply for the province itself, given the relatively limited solar and wind resources in the Sichuan Basin. Other substantial hydropower concentrations include Yunnan (84 GW, 432 TWh per year), Qinghai (47 GW, 182 TWh per year), and Hubei (35 GW, 176 TWh per year).

We analyze the benefits of nationwide cascaded hydropower optimization for capacity expansion and operation by comparing two scenarios: Scenario III and Scenario IV. The difference between these scenarios only lies in the hydropower optimization method; Scenario III employs a cascade-based optimization, whereas Scenario IV optimizes at the dam-site level. All other modeling assumptions and parameters remain identical in these scenarios. The results reveal a significant advantage to the nationwide cascaded optimization approach, which yields an annual saving of around 116 billion yuan, or 1.79% of system-wide cost (see Fig. 5B and SI Fig. S22a for cost savings decomposition). To uncover the mechanism behind this substantial saving, we investigated how coordinated dispatch reshapes the national generation mix and energy flow. We find that utility-scale solar PV and hydropower generation are higher in Scenario III than in Scenario IV, while offshore wind, biomass, and gas power generation are lower (SI Fig. S22b). Despite the increase in hydropower output, this cost reduction does not appear to stem from an increase in hydropower generation, as the difference in the national average annual capacity factor for hydropower between the two scenarios is less than 0.1%. The enhancement of system flexibility is the primary driver through which cascaded hydropower optimization

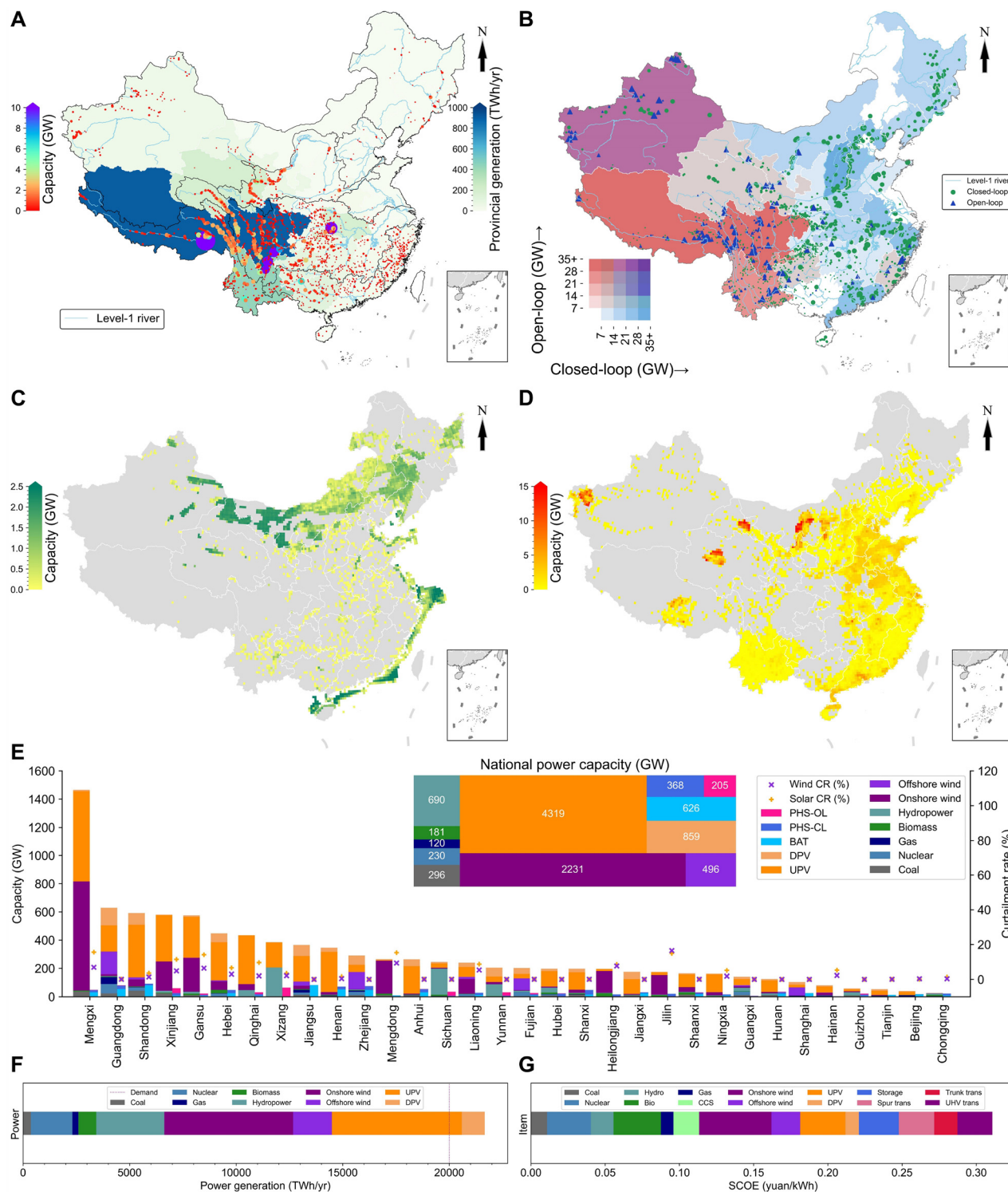


Fig. 3 Results of the base scenario for China's power systems in 2060. (A) The dam-site level installed capacity and the provincial level annual generation of hydropower. (B) Installed capacity of pumped storage (open-loop and closed-loop). (C) and (D) Cell-level installed capacity of wind power (C, onshore + offshore) and solar PV (D, utility-scale and distributed). (E) National and provincial installed capacity mix and curtailment rate (CR, %) of wind and solar power. (F) National annual generation. (G) System cost of electricity.

reduces operating costs. This flexibility, achieved by coordinating storage across a portfolio of reservoirs, allows for more power generation during periods of peak net load. The intrinsic

source of this flexibility is the capacity for seasonal arbitrage—storing water during the wet season for use in the dry season, where the net electricity load (system load minus output from

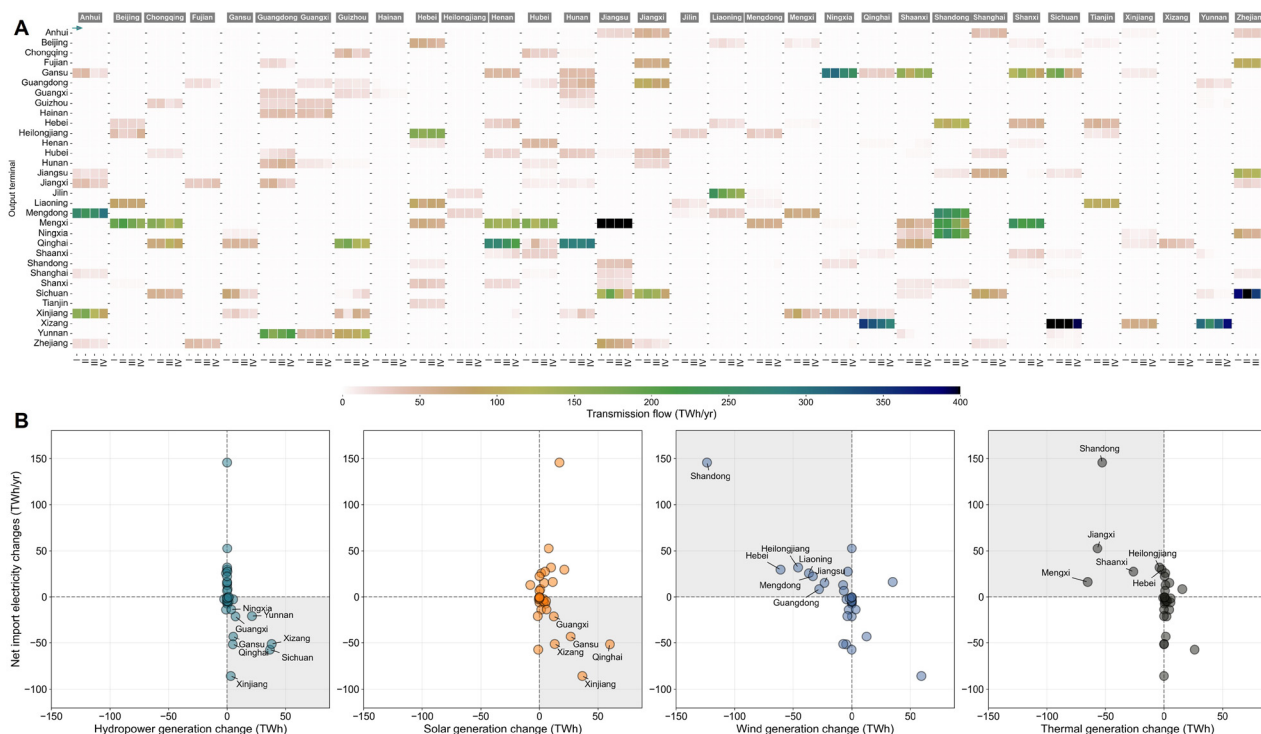


Fig. 4 Electricity flow between provinces across scenarios and generation changes in Scenario III and IV. (A) Annual electricity flow alongside transmission lines between provinces. (B) Net import electricity change and generation change in hydropower, wind, solar, and thermal power. The change is defined as the values in Scenario III minus Scenario IV.

wind and solar power) often peaks during the winter dry season at levels higher than that in summer due to reduced solar generation in winter (SI Fig. S25).

Leveraging the flexibility of hydropower cascades demands inter-regional coordination with a diverse generation mix. We find that the enhanced system flexibility afforded by cascaded hydropower provides the necessary regulatory capacity to facilitate larger-scale integration of solar PV, which, in conjunction with long-distance transmission, improves overall system economics. For instance, in the scenario featuring cascaded optimization (Scenario III), the system accommodates an additional 172 GW (+3.3%) of solar PV capacity compared to Scenario IV, despite a 16.5% reduction in flexible thermal generation (gas and biomass) and nearly identical energy storage configurations (8416 GWh vs. 8344 GWh). This new solar PV capacity is concentrated in regions such as Xizang, Guangxi, Gansu, Qinghai, and Xinjiang, which feature both abundant solar irradiance and proximity to major hydropower resources. Cascaded hydro scheduling enhances solar PV integration by creating diurnal operational flexibility with higher adjustable reservoir capacity for optimization-storing water during midday solar peaks and increasing generation at night, thus promoting solar PV deployment without significant additional investment in energy storage (see SI Fig. S26). Furthermore, this optimization paradigm boosts the output of medium-sized reservoirs (SI Fig. S24), improving their economic viability and leading to a 126 TWh per year generation increase of hydropower capacity, primarily in Xizang, Sichuan,

and Xinjiang (Fig. 4B). The economic value is ultimately realized by transmitting this low-cost power to eastern and central load centers (e.g., Sichuan to Jiangsu, Sichuan to Zhejiang, and Xinjiang to Anhui in Fig. 4A). This reduces system-wide costs by displacing more expensive local resources (e.g., offshore wind in Jiangsu) and obviating the need for renewable energy investments and associated flexible resources (e.g., biomass) in other regions transmitting electricity out, like North and Northeast China.

Beyond carbon mitigation, the hydro-dominant pathway strengthens climate adaptation and social welfare. In terms of flood resilience, national cascade optimization reduces mean peak flows by 35–40% during maximum inflow events, significantly outperforming non-cascade operations (SI Fig. S43), confirming that energy flexibility is compatible with flood control. Regarding water security, the reservoir network provides a buffer of 34.5–38.0 km³, theoretically satisfying over 10% of China's agricultural water demand, with coverage exceeding 40% in water-rich southwestern provinces (SI Fig. S44). Furthermore, the expansion of pumped storage enhances public well-being; approximately 190 (702) million people would reside within 20 (50) km of PHS stations, suggesting that these assets could serve as energy infrastructure and have potential to be recreational resources at the same time,^{35,36} particularly in densely populated coastal regions (SI Fig. S45).

We also assess the basin-scale cumulative impacts to ensure sustainable development. While prioritizing cost-effectiveness, the model internalizes environmental externalities, leading to a

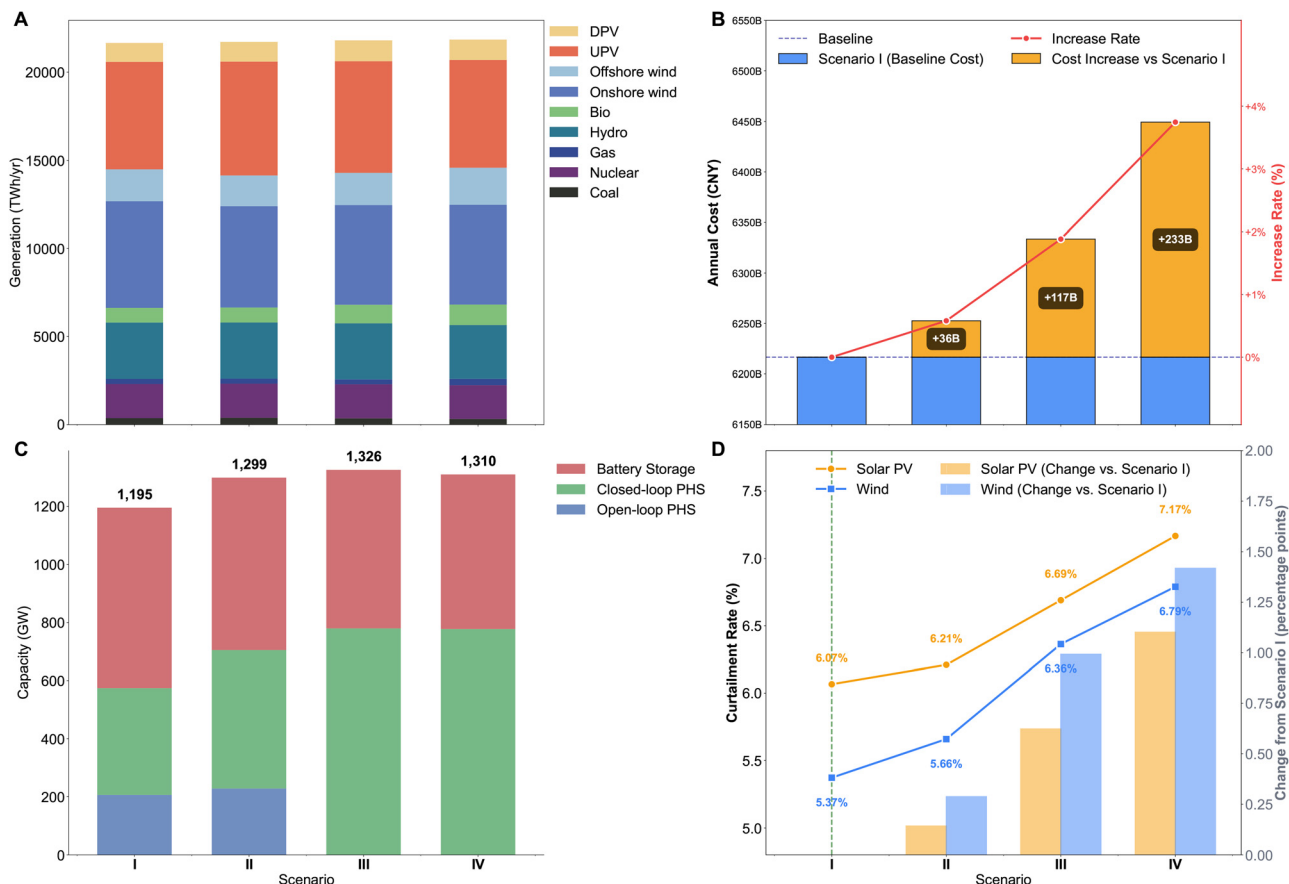


Fig. 5 Comparison of system-level results across the four modeling scenarios. (A) Annual electricity generation mix (TWh per year) by technology. (B) Changes in the total annual system cost. (C) The installed capacity mix (GW) of different energy storage technologies (battery storage, closed-loop PHS, and open-loop PHS). (D) Curtailment rates (%) for wind and solar PV power in each scenario, with bars indicating the change in curtailment compared to Scenario I.

portfolio that mitigates aggregated footprints. The deployment of the identified economically viable potential hydropower plants entails displacing approximately 106 000 people and inundating 181 000 hectares of land, with associated cumulative environmental costs estimated at 222 billion yuan (incorporated into the CISPO objective function). Spatially, impacts are concentrated in the Yangtze and Southwest River basins. However, when normalized per unit of electricity generation, these impacts remain comparatively low due to the high energy density of large-scale cascade projects. This demonstrates that by explicitly valuing social and environmental constraints during capacity planning, significant decarbonization can be achieved while maintaining cumulative basin-level impacts within manageable limits.

2.3 Role of pumped storage for decarbonizing power grids

In China, pumped hydro storage is widely recognized as a cornerstone for stabilizing the power system, with 62 GW of capacity already operational and a national target of 120 GW by 2030.⁶ Our base scenario projects a total of 368 GW of closed-loop PHS, implying that capacity must more than double in the three decades following 2030. Deployment is geographically uneven, with major concentrations in Shanxi (32 GW), Zhejiang

(27 GW), and Guangdong (24 GW). On average, these plants undergo 35 charge–discharge cycles per year, but in resource-rich regions with high shares of wind and solar, utilization is markedly higher. For example, facilities in Ningxia, Xinjiang, and Hebei operate at over 299 cycles annually, underscoring the critical role of PHS in integrating variable renewables.

In previous power system expansion models, PHS was represented at the grid-regional level with a fixed storage duration, typically 8 hours, and in some cases up to 10 hours (e.g., ref. 2, 20, 21, 27 and 28). In this study, we model closed-loop PHS at the dam-site level, allowing storage duration to vary by site. This approach yields a mean storage duration of 16 hours across 1400 optimized sites, with the maximum duration of over 300 hours. The comparison indicates that fixed-duration representations are unable to capture the spatial heterogeneity of PHS resources. To assess the implications of this difference, we constructed Scenario II, in which PHS expansion and dispatch are modeled at the provincial level with a fixed 8-hour duration. Compared with Scenario I (site-level modeling), Scenario II results in a 36 billion yuan per year (+0.6%) increase in total system cost (see Fig. 5B). Installed closed-loop PHS capacity increases by 30% to 478 GW (Fig. 5C), while wind and solar curtailment rates increase by 5.4% and 2.4%,

respectively. Open-loop PHS capacity also rises by 11%. These differences are attributable to the reduced accuracy in representing closed-loop PHS duration and flexibility when using grid-level, fixed-duration assumptions.

In addition to closed-loop PHS, open-loop PHS has the potential to supplement China's energy storage infrastructure. However, previous national studies have not incorporated open-loop PHS, which needs an explicit coupling with hydropower at the dam-site level. Here, we co-optimize the capacity expansion and hourly dispatch of open-loop PHS in conjunction with all other generation units. In our modeling framework, open-loop PHS is configured as a hybrid system capable of sharing a reservoir with a medium- or large-scale hydropower plant and drawing electricity from the grid for pumping. Our optimization results indicate a national requirement of 205 GW of open-loop PHS by 2060, with a geographical distribution that closely mirrors that of major hydropower resources, as depicted in Fig. 3B. The leading provinces include Xizang (62 GW), Xinjiang (35 GW), Sichuan (30 GW), Yunnan (24 GW), and Qinghai (13 GW).

Open-loop PHS can increase system flexibility with only minimal resource requirements. Comparing scenarios II (with open-loop PHS) and III (without open-loop PHS), we find that deploying open-loop PHS in China's high-renewable power system yields annual cost savings of 81 billion yuan ($\sim 1.3\%$), see Fig. 5B. These savings primarily stem from enhanced hydropower regulation, which improves system-level dispatchability; for instance, compared with Scenario III, annual hydropower generation only increases by 0.27% (~ 8.5 TWh per year), *i.e.*, not from higher hydropower generation. This flexibility further reduces wind and solar curtailment by 11% and 7.2%, respectively, see Fig. 5D. Remarkably, these gains require only ~ 3942 km³ of additional reservoir capacity—less than one-tenth of the Three Gorges reservoir. By contrast (comparison between scenarios II and III), in a net-zero system without open-loop PHS, far greater low-carbon firm capacity is needed: bioenergy generation rises by 24% (+205 TWh per year, Fig. 5A), and closed-loop PHS capacity expands by 64% to 780 GW (Fig. 5C). Altogether, these results underscore the pivotal role of open-loop PHS in enabling a cost-effective, flexible, and low-carbon electricity system.

2.4 Price volatility dynamics and the economic viability of storage

The economic viability and operational stability of future renewable-dominated power systems hinge on sufficient flexibility to balance variable generation with demand. The distribution of the marginal cost of electricity supply serves as a significant diagnostic for system status, where the cost reflects the price of supplying one extra unit of energy when power demand is balanced.³⁷ In systems with high shares of zero-marginal-cost renewables, the emergence of zero or negative prices is a key indicator of generation surplus coupled with system inflexibility. A negative price signals that generators are willing to pay to offload power to avoid costly curtailment or shutdowns, revealing a temporary inability of the grid to either

store or transmit the surplus energy. Our analysis of marginal cost distributions in China's 2060 power systems under different modeling frameworks reveals that the representation of flexibility resources is non-trivial, but an important driver of projected market outcomes. Our base scenario (Scenario I) incorporates a detailed representation of key flexibility mechanisms from hydro, including nation-scale cascade hydropower operations and the inclusion of both open- and closed-loop pumped storage. As illustrated in Fig. 6, the resulting price distributions in this scenario show a system with better-managed volatility, compared to other scenarios with limited representation of these hydro-based flexibilities. Across most regions, the median marginal price is near zero, reflecting the dominance of renewable resources, while a long right-side tail indicates periods of scarcity that are met by high-value flexibility from energy storage and dispatchable sources. The frequency and magnitude of negative prices are much better contained. This outcome demonstrates that a system equipped with a robust and geographically diverse portfolio of flexibility options, particularly the large-scale energy shifting capability of cascade hydro and PHS, can effectively integrate most renewable generation surpluses, thereby stabilizing prices and lowering curtailment.

The critical importance of accurately modeling these flexibility resources becomes starkly evident when their representations are simplified. Scenario III, which optimizes the operation of closed-loop PHS at the grid level and does not represent open-loop PHS, reveals a dramatic shift in market dynamics compared to the base scenario (Scenario I). The system's ability to integrate low-cost surplus generation is constrained, resulting in increases in both the frequency and depth of negative prices across almost all regions. This effect is further amplified in Scenario IV, which not only excludes open-loop PHS but also simplifies hydropower modeling from an interconnected cascade to isolated, dam-level operations. This simplification erases the inherent storage and operational coordination capacity of entire river systems. Consequently, Scenario IV, lacking both open-loop PHS and cascade coordination, projects the most extreme price volatility, with the most frequent and severely negative prices. The mechanism is clear: as the model is stripped of flexibility options, it is forced to balance the grid through the only remaining lever—massive curtailment of renewable energy.

These pronounced price dynamics have profound implications for the financing of pumped storage. In China, while tens of gigawatts of PHS have been deployed, their economic viability is currently constrained by a near-flat electricity price system that has minimal price volatility, thereby preventing revenue generation through energy arbitrage. To overcome this, authorities established a regulated capacity payment mechanism, providing PHS plants with a fixed annual revenue based on their capital cost to ensure investment recovery.³⁸ In this study, we evaluate the model-optimized closed-loop PHS at the site level using the current designed capacity payment mechanism to ensure annual capital recovery, exclusive of revenues from electricity market arbitrage. We find that this

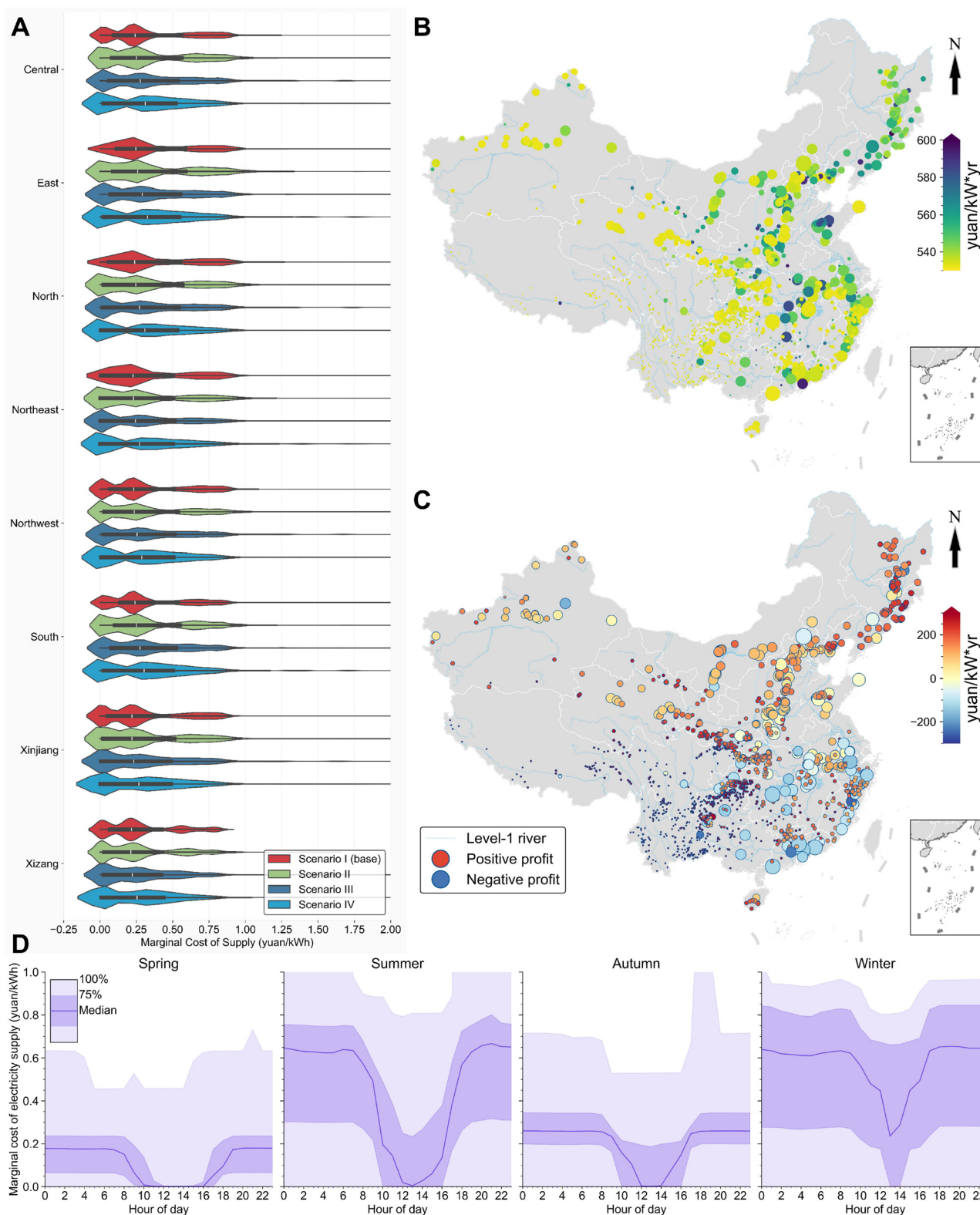


Fig. 6 Distribution of marginal cost of electricity supply and capacity price for closed-loop pumped storage. (A) Violin plots for the marginal cost of electricity supply (yuan per kWh) in each power grid region across scenarios. (B) Capacity price (yuan per kW year) for closed-loop pumped storage using the current designed subsidy mechanism. (C) Annual profits (yuan per kW year) for closed-loop pumped storage considering revenue from the electricity market. (D) Hourly marginal cost distribution of each season in the base scenario.

administratively determined subsidy, as shown in Fig. 6B, results in significant required capacity payments (500–650 yuan per kW year) across the country, representing a substantial and sustained financial commitment from the state. This approach, while necessary in a regulated environment, reflects a potentially inefficient, out-of-market solution.

Fundamental transformation, driven by market-oriented reforms such as the policy stipulated in Document No. 136 by the National Development and Reform Commission and the National Energy Administration in 2025,³⁹ starts to reshape the landscape for grid-scale storage. The marginal cost of electricity supply in a power system with high renewable penetration, as simulated in our base scenario, inherently reflects the price signals that are previously absent. Fig. 6D illustrates this emergent market dynamic, revealing pronounced diurnal marginal price volatility across all seasons. In summer, for instance, prices plummet to near zero during midday hours due to abundant solar generation, before soaring to over 0.6 yuan per kWh during the evening peak as demand outstrips solar output. This predictable and significant daily price spread creates a previously unavailable revenue stream for PHS through energy arbitrage—charging with cheap, surplus electricity and discharging expensive, high-demand electricity.

This market-driven revenue fundamentally alters the economics of PHS. Based on the hourly provincial shadow prices of the electricity demand constraint (*i.e.*, marginal cost of electricity supply) and the charge/discharge volumes of each closed loop PHS site reported by the CISPO model, we calculate the annualized profit for each plant, factoring in payments for charging, revenue from discharging, and annualized investment and operation and maintenance (O&M) costs. Fig. 6C illustrates the 2060 profit status of each PHS site, where the blue spectrum represents negative profits and the red spectrum indicates positive profits. For unprofitable units, capacity payments are required to ensure their expected return on investment. The required capacity payments are drastically reduced nationwide, falling below 200 yuan per kW year in most locations. Notably, a significant number of potential PHS sites, particularly in the renewable resource-rich regions of North and Central China, now exhibit positive profits. This critical finding demonstrates that in a well-functioning electricity market, the arbitrage opportunities could be substantial, so that PHS projects can become commercially viable without any subsidy.

Decomposition of storage trajectories reveals that while large hydropower reservoirs dominate seasonal balancing, closed-loop pumped storage operates primarily on multi-day to multi-week scales (see SI Fig. S33). Within this operational window, we observe a robust relationship between storage duration and economic viability. As illustrated in Fig. 7, annual profit per installed capacity increases monotonically with designed duration: units restricted to short-term storage (<1 day) yield low or negative median profits (≈ -50 yuan per kW), whereas long-duration designs (>30 days) consistently achieve the highest profitability (>700 yuan per kW). See the SI, Fig. S34 for these results for each provincial grid. This pattern is substantiated by a strong negative correlation ($R \approx -0.86$)

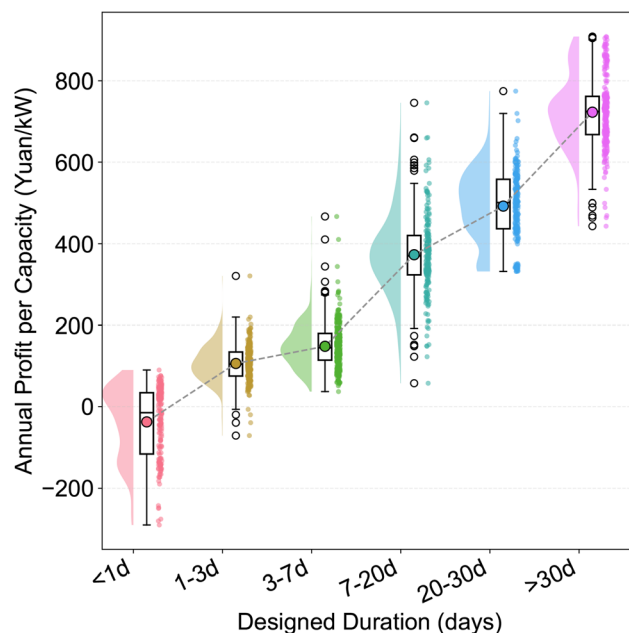


Fig. 7 Relationship between annual profit per installed capacity (yuan per kW) and designed duration (days) for all closed-loop PHS.

between profit per kW and realized cycling frequency *via* the rain-flow counting algorithm⁴⁰ (see the SI, Fig. S35). The analysis shows that the most profitable units are not those performing continuous intraday cycling, but rather those that cycle less frequently—typically tens of times per year. This indicates that in power systems characterized by high VRE penetration and scarcity pricing, unlike short-duration units constrained to daily cycles, long-duration PHS possesses the capacity to bridge extended periods with larger renewable supply differences. This economic advantage is attributed to the significantly larger electricity price differentials accessible over longer horizons. Our analysis demonstrates that while the maximum intraday price spread typically hovers around 0.25 yuan per kWh, extending the temporal window to one week can nearly double this figure.

2.5 Influence of weather conditions on the power system

Weather conditions may influence renewable energy generation, introducing uncertainties into low-carbon power systems that derive over 70% of their electricity supply from renewable sources.⁴¹ This study adopts the meteorological data in 2019 as the base weather year (including solar radiation, wind speed, and river discharge) because the only open hourly provincial demand profile is in 2019.⁴² To evaluate potential weather-related impacts on China's power systems projected for 2060, we then optimize additional scenarios using meteorological data from 2000 to 2018. The full-scale model—which incorporates national-level cascade optimization of hydropower and reservoir-level dispatch of pumped storage—requires approximately 30 days to solve our high-performance computing platform, making it difficult to test tens of scenarios. Therefore, our weather condition scenarios employ dam-site level optimization for

hydropower and power grid level modeling for pumped storage (equivalent to Scenario IV). Despite methodological differences between these approaches, the streamlined model provides valuable comparative insights across multiple weather years, yielding comparable results for key metrics including

wind and solar installation capacity, system costs, and annual generation.

Fig. 8 shows the impact of varied weather conditions on key power system characteristics, including annualized system cost, installed capacity, generation mix, and the distribution

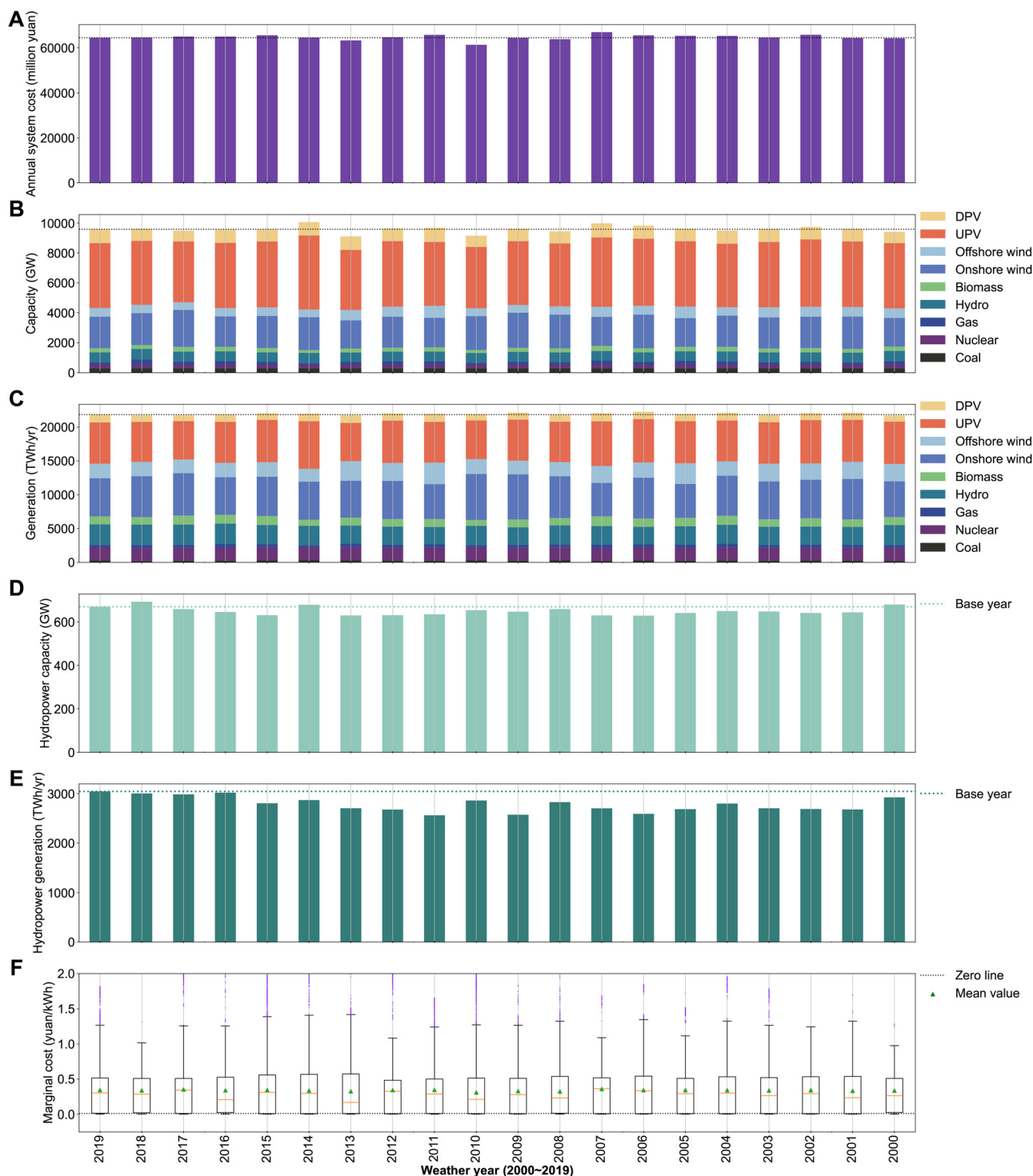


Fig. 8 System level results of different historical weather conditions (2000–2019). (A) Annual system cost (100 million yuan per year). (B) Power capacity (GW) mix. (C) Electricity generation (TWh per year) mix. (D) Installed capacity (GW) and (E) annual generation (TWh per year) of hydropower. (F) Boxplots of hourly marginal cost (yuan per kWh) of generation.

of marginal generation cost. Our analysis demonstrates that China's renewable energy-dominated power systems, designed to meet the 2060 carbon neutrality target, show variability in key metrics while maintaining consistency across scenarios. Using 2019 weather data as our baseline, the CISPO model requires 9590 GW of installed generation capacity to satisfy the projected annual power demand of 20 PWh. When tested against twenty historical weather years (2000–2019), this capacity requirement ranges from 9112 GW (−4.9%) to 10 004 GW (+4.9%), reflecting variations in renewable generation potential. The annual system cost of China's power systems optimized by the CISPO sees similar variance to the installed capacity, from 6.45 (−4.9%) to 6.70 (+3.7%) trillion yuan. In the base weather year, the 9590 GW capacity produces approximately 21.85 PWh of electricity annually, accounting for renewable curtailment, transmission losses, and storage inefficiencies. Our system-level optimization across the entire planning year shows remarkably consistent total generation, ranging narrowly from 21.75 PWh per year (−0.5%) to 22.27 PWh per year (+1.9%). The consistent surplus between generation and demand (20 PWh per year) underscores the necessity of energy balancing technologies. Optimal deployment of closed-loop PHS and BAT in the base weather year reaches 778 GW and 532 GW, respectively, with ranges of 526–836 GW for PHS and 532–665 GW for BAT across all weather scenarios.

While planning for hydropower exhibits a consistent trend across various meteorological conditions, it is crucial to recognize the inherent variations. The total installed capacity for hydropower ranges from 630 to 690 GW, with annual electricity generation falling between 2.5 and 3.0 PWh per year. There is no direct correlation between the installed hydropower capacity and the annual water availability. For instance, choosing years with abundant water resources, such as 2016, requires less capacity to meet the electricity supply. Conversely, in years with water scarcity, such as 2006, less capacity is also deployed due to the diminished generation potential of some hydropower stations. Despite the relatively stable scale of hydropower planning across different meteorological years, the systemic impacts of drought years—for example, 2006, 2011, and 2012—warrant attention. In these years, a reduction in hydropower generation coincides with more favorable conditions for wind and solar power, resulting in minimal changes to the total system cost. The meteorological conditions of 2007 represent a noteworthy exception, where poor generation conditions simultaneously apply to wind, solar, and hydropower—the three primary future renewable energy sources. This confluence of adverse conditions necessitates greater installed capacity and flexibility,⁴¹ leading to a significant increase in electricity costs (+3.9%). On the whole, the future electricity system of China demonstrates adequate resilience to face diverse meteorological conditions. Beyond techno-economic resilience, this multi-year analysis further validates the robustness of our cumulative impact assessment; normalized environmental and social indicators—such as population displacement per unit of generation—exhibit consistent magnitudes across historical

hydrological conditions (see SI Fig. S37–S42 for detailed results), confirming that the system's basin-wide footprint remains stable within optimized boundaries.

The marginal cost of electricity supply remains consistent overall across different meteorological years, yet exhibits significant disparities in its extreme values. An analysis of the distribution of hourly marginal electricity prices across all provinces for each meteorological year, as depicted in Fig. 8F, reveals this pattern. Across all years, the mean marginal price ranges from 0.31 to 0.36 yuan per kWh, the 25th percentile from 0.004 to 0.024 yuan per kWh, the median from 0.16 to 0.36 yuan per kWh, and the 75th percentile from 0.48 to 0.57 yuan per kWh. These metrics indicate a generally consistent distribution of marginal costs of electricity supply. In stark contrast, the maximum intra-annual marginal price varies dramatically. For instance, the lowest maximum price is 3.0 yuan per kWh, occurring in 2008, whereas extremely high maximum marginal prices (tens to hundreds yuan per kWh) are also observed in some weather years. For the base year of 2019, the maximum price is 65 yuan per kWh (for specific values, see SI Table S5). These findings suggest that the marginal cost of electricity supply in a future renewable-dominant electricity system remains broadly stable within an expected range. However, extremely high values, typically driven by short-term, extreme meteorological events, necessitate careful attention. Furthermore, negative electricity prices are observed in all meteorological years.

3 Discussion

In this study, we demonstrate that the systemic value of accurately modeling pumped storage and hydropower is a critical, yet frequently underestimated, determinant of an economically efficient decarbonization pathway by developing the China Integrated Sustainable Power-system Optimization (CISPO) model to simultaneously simulate the capacity expansion and cascade-based operation for hydropower and pumped storage at the reservoir site level for the first time. Our high-fidelity, nation-scale optimization of China's 2060 power system reveals that the economic value of cascaded hydropower operation amounting to an annual system-wide cost saving of 116 billion yuan is driven not by the increase in total hydropower generation but by enhanced seasonal flexibility—storing water during wet seasons to supply power during high-net-load winter periods, which in turn enables more integration of low-cost solar PV. Furthermore, we find that the significant diurnal price volatility inherent in a renewable-dominated grid creates a powerful market-based revenue stream for pumped storage through energy arbitrage, potentially rendering many new projects commercially viable without government subsidies and drastically reducing the required capacity payments for others. While our multi-year weather analysis confirms the system's overall resilience, it also highlights that rare, extreme meteorological situations impacting wind, solar, and hydro resources can lead to significant cost spikes, underscoring

the necessity of meticulous planning for backup capacity. Supplementary analysis also indicates substantial non-energy dividends: the optimized hydropower and PHS fleet could provide a mean peak flow reduction of 35–40% for flood mitigation, satisfy approximately 11% of the national agricultural water demand, and place over 700 million people within recreational reach of pumped storage facilities. These findings have several policy implications for both near- and long-term planning.

First, our findings suggest that while the global energy transition narrative is rightly dominated by the exponential growth of wind and solar power, this focus risks marginalizing the equally critical and monumental role of hydropower resources in achieving deep decarbonization. We reveal that China possesses a total technical hydropower potential of around 715 GW, a capacity nearly double its current installation and concentrated predominantly in its southwestern regions, alongside a vast and largely untapped reservoir of energy storage potential in the form of 390 GW of open-loop PHS, which can be integrated with hydropower facilities, and 1960 GW of closed-loop PHS. Critically, our optimization for a 2060 carbon-neutral system demonstrates that a cost-effective solution necessitates deploying a significant fraction of this capacity, approximately 690 GW of hydropower, to serve as a cornerstone of clean, dispatchable energy, complemented by 205 GW of open-loop and 368 GW of closed-loop systems to provide the long-duration storage essential for stabilizing a grid dominated by variable renewables. The systemic value of this large-scale hydropower deployment is unlocked through cascade optimization, which extends well beyond conventional water resource efficiency. By coordinating reservoir storage across river basins, this approach provides enhanced seasonal flexibility that strategically matches supply with demand during critical winter periods when solar generation is constrained. This coordinated operation transforms system planning economics by enabling higher-scale solar PV deployment while reducing dependence on expensive flexible thermal generation. Therefore, the full value of these resources emerges from the synergistic properties of interconnected hydro-storage networks, demanding a national energy strategy that evolves beyond a singular focus on VRE expansion to embrace systemic thinking in both the planning and operation of these foundational flexibility assets.

Second, this study's high-fidelity approach further reveals that the high-granular modeling of system flexibility is not merely an academic exercise but a decisive factor in shaping projected market outcomes and designing effective policy. Our analysis demonstrates that simplified models may constrain flexibility, which can lead to inefficient policy decisions. For instance, the current reliance on administratively set capacity payments to ensure the viability of pumped storage is a tool designed for a regulated environment with relatively flat prices. By resolving PHS investment and hourly operation at the individual reservoir level, our model transcends such aggregate assumptions to precisely quantify the economic prospects of each potential plant within a dynamic, market-driven context.

This granular analysis yields an economic insight: the pronounced diurnal price spreads naturally emerging in a VRE-dominated system create abundant energy arbitrage opportunities that alter the need for subsidies. Our results show that required capacity payments would be drastically reduced nationwide, falling below 200 yuan per kW year in most locations, and, most critically, that a significant number of PHS projects in renewable-intensive regions of North and Central China could become commercially profitable through market participation alone. This suggests that instead of heavily relying on out-of-market subsidy mechanisms, unlocking the massive potential of grid-scale storage should accelerate the development of robust electricity markets where price signals accurately reflect the intrinsic value of flexibility. The site-level modeling framework demonstrates that the “marketplace for storage” reveals that the economic viability of storage is location-dependent and driven by nodal price volatility rather than uniform national policies. By capturing these granular market dynamics, our approach provides a generalizable blueprint for other nations to identify where and how storage technologies can transition from subsidy reliance to market-driven profitability.

Our results define a techno-economic optimum; however, realizing this potential requires overcoming significant institutional and governance barriers. Our analysis demonstrates the economic stakes of these non-technical factors: the 116 billion yuan annual cost difference between Scenarios III and IV represents the opportunity cost of fragmented governance. Currently, major river basins (*e.g.*, the Yangtze) are managed through a mix of provincial and corporate stakeholders with competing objectives—flood control, irrigation, and maximization of local generation. Transitioning to the “integrated governance” paradigm modeled in the Base Scenario requires establishing robust *trans*-regional management authorities capable of aligning these diverse interests. Furthermore, policy evolution is a prerequisite for fully realizing the economic value of pumped storage. As our marginal cost analysis indicates (Fig. 6), the profitability of PHS hinges on arbitrage within volatile markets. Therefore, the critical policy evolution required is not merely the expansion of subsidies, but the acceleration of spot market liberalization that monetizes flexibility. Regarding environmental permitting, while our model strictly excludes ecologically sensitive zones, procedural hurdles remain. The complexity of land acquisition and environmental impact assessments for gigawatt-scale projects may extend lead times, potentially constraining the deployment pace. These findings have broad global transferability. The “institutional friction” we have identified in China mirrors challenges in other major transboundary basins, such as the Mekong, Nile, or Amazon. For nations sharing these hydrological resources, our results demonstrate that failing to establish multi-stakeholder coordination mechanisms entails a substantial economic cost in the low-carbon transition.

Third, the multi-year meteorological analysis spanning two decades reveals a great challenge of the inherent unpredictability of extreme multi-resource stress events confronting

carbon-neutral electricity systems. While aggregate system metrics exhibit remarkable consistency, with total generation capacity requirements varying only $\pm 4.9\%$ and costs fluctuating within narrow bands, this apparent stability masks critical vulnerabilities during extreme weather events. Under these conditions, simultaneous stress on wind, solar, and hydropower resources triggers substantial spikes of marginal electricity supply cost spanning three orders of magnitude. The confluence of poor wind, solar, and hydropower conditions observed in 2007 exemplifies scenarios that necessitate proactive planning frameworks explicitly accounting for tail-risk events. Such frameworks must maintain adequate backup capacity to prevent catastrophic system failures that could undermine entire decarbonization pathways. The implications extend beyond technical optimization to encompass strategic policy considerations for resilient renewable energy transitions.

The core contribution of this paper is the systematic evaluation of the hydropower's role (including open- and closed-loop pumped storage) in decarbonizing the power system on a continental scale (*i.e.*, China) using an ultra-high spatial and temporal resolution planning model for the first time. Within this defined research boundary, we acknowledge several limitations, although we believe they do not affect our main findings. First, to maintain focus on the unprecedented spatiotemporal modeling of wind, solar, hydro, and pumped storage, we simplify certain nonlinearities. These include the dynamic water head simulation of hydropower and the unit commitment using integer programming for thermal and nuclear power. Second, as our analysis centers on the intrinsic impact mechanisms of hydropower modeling, we adopt widely accepted socioeconomic parameters from previous research, such as electricity demand profiles, cost assumptions, and carbon emissions constraints, without conducting additional uncertainty analysis on these variables, which has been done in existing research.^{2,4,20,26,43} Third, we do not model several non-primary electricity generation technologies, including concentrating solar power, geothermal energy, and offshore photovoltaics. Fourth, the potential impacts of future climate change are not considered. This is mainly because the precision of climate variables from current global climate models under different pathways is insufficient to support our high-resolution spatiotemporal modeling; for instance, crucial data like solar radiation are often only available at a coarse $1^\circ \times 1^\circ$ resolution and on a sub-daily basis. Fifth, our assessment of hydropower flexibility, while considering environmental flows and reservoir volume constraints, does not include dynamic operational rules for non-power objectives, such as flood control, which can limit real-world reservoir flexibility. Consequently, the quantified seasonal flexibility and its associated ¥116 billion annual saving from cascade optimization should be interpreted as a techno-economic upper bound, with the realizable value in practice likely being lower due to such operational constraints.

Building on this work, future research could proceed in several key directions. First, future modeling could incorporate forward-looking climate scenarios to assess system resilience and optimal configurations under climate change.⁴⁴ Second,

the CISPO framework can be extended to offer a novel integrated modeling approach for the water–energy nexus, enhancing the representations of other hydro operation objectives, including flood control, drought relief, *etc.*,⁴⁵ to enable a quantitative analysis of the trade-offs and synergies between energy security and water security. Third, research is needed to design electricity market products and rules that can efficiently price flexibility while protecting consumers from extreme price volatility. Fourth, the CISPO's *trans*-basin hydropower system modeling can be extended to assess the potential decarbonization of inland waterway transportation through electrification. Fifth, developing comprehensive economic valuation frameworks for non-energy services would enable integrated full-spectrum cost-benefit analyses for hydropower and pumped storage.

In summary, the CISPO model provides a synergistic optimization of wind, solar, hydropower, and pumped hydro storage that offers new insights for power sector transitions. The insights from this study, while focused on China, have global relevance. China's hydro-centric flexibility strategy offers a powerful model for nations with similar resource endowments (*e.g.*, in South America, North America, or parts of Africa). For countries lacking large-scale hydropower potential, our findings highlight the urgent need to invest in alternative sources of long-duration storage or to prioritize robust cross-border grid interconnections. This integrated methodology establishes a useful tool for assessing decarbonization pathways, emphasizing that the future of renewable energy is inextricably linked to the intelligent and systemic management of water.

4 Materials and methods

4.1 Assessment of hydropower and pumped storage

To model hydropower and pumped storage at the reservoir site level, numerous attributes (*e.g.*, hydraulic head height, reservoir capacity, and power capacity) of existing and potential plants are fundamental input data to the CISPO. In this study, we first collect the information on installed, under-construction, and in-plan hydropower and pumped storage from multiple-source datasets and then merge their attributes according to the location of each plant. For the installation potential of new plants, we conduct a high-resolution resource assessment for hydropower and pumped storage, considering geographical conditions, environmental protection, cost effectiveness, social impacts, and overlap with existing installed capacities.

4.1.1 Assessment of hydropower. We collect and merge existing hydropower attributes from the Chinese Hydropower Database (CHD),^{46,47} the Global Hydropower Tracker (GHT),⁴⁸ and the Large Dam Safety Supervision Center (LDSSC)⁴⁹ of the National Energy Administration, China, which provides detailed technical and geographical parameters. The parameters include installed capacity (MW), completion years, reservoir capacities (m^3), spillage outflow capacities ($\text{m}^3 \text{s}^{-1}$), and hydraulic heads (m). To compensate for the limited availability of real-time dispatch manuals, we approximated

non-power operational constraints (e.g., flood control, irrigation, and ecological protection) by integrating administrative regulatory data with rigorous physical boundaries. First, we utilized the LDSSC database⁴⁹ as the primary source for reservoir parameters. We strictly constrained the operation of existing plants within their officially registered Normal Storage Level and Dead Storage Level. These regulatory bounds are legally established to reserve storage capacity for flood buffers and downstream water supply, thereby acting as a robust proxy for non-power service requirements. Crucially, to ensure future projects adhere to these same standards, we derived a regulating-to-normal storage capacity ratio (0.470) from the LDSSC records and applied this empirical constraint to the design of all potential hydropower sites (see SI Section 2.1 and Fig. S7). Second, environmental flow requirements were enforced as a hard constraint ($q_{g,i}^{\text{hp,env}}$), defined as the 30th percentile of historical daily discharge (Q_{30}) over a 40-year period (1980–2019).¹⁷ This ensures ecological baselines are met prior to power generation. Third, the physical characteristics of cascades were represented using dynamic hydraulic delays. We employed cross-correlation function (CCF) analysis on historical streamflow data to calculate precise water travel times (τ) between reservoirs, ensuring that cascade operations respect the physical inertia of water mass transfer (see SI Section 2.3 and Fig. S15).

For the installation capacity potential of newly built hydropower, we adopt virtual dam-site-based assessment methods.^{16,17} First, we set virtual dam-sites for the hydropower project alongside the river network in China at each 4.5 km, which totals about 72 000 virtual sites. We then develop a parallel recognition algorithm to generate inundation areas for each virtual dam site with the dam height ranging from 10 to 300 m, using a high-resolution DEM dataset.³¹ The dam height of the virtual site will first be excluded if the inundation area overlaps with installed hydropower, historical landmarks, or protected zones (e.g., environmental and ecological conservation areas). For the remaining potential dam height assumptions, we estimate their life cycle levelized cost of electricity (LCOE), considering engineering, population relocation, land use acquisition, and equipment cost, with generation potential assessed by discharge data.³² Subsequently, we select the dam height assumption with the least LCOE for the installation capacity potential, reservoir capacity, and other key parameters. With the upper bound of LCOE at 0.5 yuan per kWh, we find a total of about 715 GW of hydropower potential with existing capacity in China. See the detailed procedure and technical-economic parameters used for hydropower resource assessment in the SI Section S2.

4.1.2 Assessment of pumped storage. This study quantifies the technical potential of open- and closed-loop PHS in China through a multi-stage techno-economic assessment. Our methodology is founded upon a global atlas of 616 000 potential PHS sites,¹⁸ which we integrate with empirical data for existing projects from the CHD⁴⁶ and GHT⁴⁸ databases. We evaluate two distinct PHS typologies: hybrid open-loop systems, whose configurations are optimized by pairing new reservoirs with hydropower dams, and standalone closed-loop systems. For both

typologies, the design of each potential site is optimized to minimize its levelized cost of storage (LCOS). A rigorous screening filter is then applied to all sites to resolve spatial conflicts with protected areas and critical infrastructure and to enforce environmental and economic viability criteria. The analysis identifies a national potential of approximately 390 GW (open-loop) and 1960 GW (closed-loop) under an LCOS constraint of 0.25 yuan per kWh, with procedural details available in SI Section S3.

4.2 Integrated sustainable power-system optimization model

In this study, we substantially enhance the CISPO Model,⁴ a sophisticated power system expansion framework characterized by exceptional spatial and temporal granularity, through the introduction of novel datasets and constraints specifically designed to model hydropower and pumped storage systems (encompassing both open-loop and closed-loop). Aligned with prominent capacity expansion models in the literature,^{1,2,5,20,22,50} the CISPO model's primary optimization objective centers on annualized system costs, which incorporate capital expenditures and operational expenses. To effectively capture the intricate spatial heterogeneity and pronounced seasonal variability of renewable energy resources, the model co-optimizes capacity expansion and system operation at an unprecedented hourly resolution across a complete planning year (8760 h), with VRE capacity decisions localized at a grid-cell level ($0.25^\circ \times 0.25^\circ$, about 25 km $\times$ 25 km at mid-latitude). The optimization outputs of the CISPO model include capacity deployment for generation, storage, inter-provincial transmission, and carbon capture and transportation infrastructure, as well as hourly commitment, dispatch, and electricity flow decisions throughout the planning horizon. The model's default constraint framework encompasses a robust set of considerations, including capacity expansion limits for resource-constrained technologies, hourly operational limits, unit-commitment for thermal and nuclear power, emission targets, carbon capture and sequestration limits, inter-provincial transmission power flow, and system security requirements such as reserve and inertial response capabilities. As a cutting-edge system expansion model distinguished by its high spatial resolution and extensive nationwide temporal coverage, the CISPO framework here is uniquely positioned to simulate the systemic value of hydropower and pumped storage technologies within the context of a deeply decarbonized power system. For methodological clarity, we focus on presenting the enhanced hydroelectric and storage-related components of the CISPO model, with the understanding that all other constraints and parameters remain consistent with its previous version as described in ref. 4.

4.2.1 Objective function. To more accurately capture the spatial heterogeneity in investment costs for hydropower and pumped hydro storage, including both open-loop and closed-loop systems, across different locations, we incorporate site-specific capital expenditure (CapEx) assessments into the objective function of the CISPO framework. The updated objective function is expressed as follows:

$$\min : f = f^* + \sum C_{g,i}^{\theta}, \quad \forall g \in G, i \in I, \theta \in \Theta, \quad (1)$$

where f^* represents the system cost optimized in ref. 4, excluding hydropower and PHS components; $C_{g,i}^\theta$ denotes the annualized cost of the i -th generation technology θ within provincial grid g , encompassing site-specific annualized capital expenditures (accounting for engineering and environmental factors), annual fixed operation and maintenance (O&M) costs, the aggregated hourly variable O&M costs, and distance-related intra-provincial transmission costs (first to the substation for all technologies and subsequently to the load center for hydropower). Here, G represents the set of power grids (32 provincial grids in mainland China), I denotes the set of site-specific generation technologies, and Θ encompasses hydropower, open-loop PHS, and closed-loop PHS. In our multi-scenario analysis, if a specific generation technology is not modeled (e.g., open-loop PHS), its associated costs are excluded from the system objective. Additionally, if the technology is optimized at the power grid level (e.g., closed-loop PHS), the capital expenditure is scaled up proportionally based on the installation capacity potential, calculated as $\sum \text{Cap}_{g,i}^\theta \times \text{CapEx}_{g,i}^\theta / \sum \text{Cap}_{g,i}^\theta$, ensuring a comprehensive representation of regional investment dynamics.

4.2.2 Constraints on output of hydropower. For each hydropower plant i in grid g , its power output $\text{inte}_{g,i,t}^{\text{hp}}$ at time t is determined by the hydraulic head ($H_{g,i}^{\text{hp}}$, m) and the discharge flowing through turbines (generation flow, $q_{g,i,t}^{\text{hp,gen}}$, $\text{m}^3 \text{s}^{-1}$) according to:

$$\text{inte}_{g,i,t}^{\text{hp}} = q_{g,i,t}^{\text{hp,gen}} \times H_{g,i}^{\text{hp}} \times \eta_{\text{hp}} \times \rho \times g_e, \quad \forall g, i, t, \quad (2)$$

where η_{hp} is the output efficiency of hydropower (0.85 used), ρ is the density of water (1000 kg m^{-3}), and g_e is the gravitational acceleration (9.8 m s^{-2}). Because this study allows hydropower to expand its capacity, the power output is also constrained by the installation capacity. Additionally, hydropower can provide security capacity, such as inertia requirements and spinning reserve. Inertia from hydropower is the product of installed capacity and the inertia constant and is added into the inertia requirement constraint in ref. 4. Constraints on power output and reserve spinning are as follows:

$$\text{inte}_{g,i,t}^{\text{hp}} \leq z_{g,i}^{\text{hp}} \times \bar{p}_{g,i}^{\text{hp}}, \quad \forall g, i, t, \quad (3)$$

$$\text{res}_{g,i,t}^{\text{hp,+}} \leq z_{g,i}^{\text{hp}} \times \bar{p}_{g,i}^{\text{hp}} - \text{inte}_{g,i,t}^{\text{hp}}, \quad (4)$$

where $\bar{p}_{g,i}^{\text{hp}}$ is the installation capacity potential assessed before the optimization, $z_{g,i}^{\text{hp}} \in [0,1]$ is a decision variable on the installation factor of hydropower i in planning year, and $\text{res}_{g,i,t}^{\text{hp,+}}$ is the spinning reserve provided by hydropower i at time t . In this study, we model the small hydropower (about 22.8 GW in total), with installed capacity less than 50 MW according to the official Chinese industry standard for hydropower project rank classification³³, at the provincial grid level to accelerate computing, as their capacity is limited (less than 3% of total potential). In this modeling method, the reservoir of each small hydropower is summed to be a provincial storage equipment, and the natural water inflow with the hydraulic

head calculated using equation 2 is treated like charging in energy storage modeling.^{2,4,20} The electricity to the grid from these small hydropower plants is subtracted from the “equivalent” provincial storage capacity.

4.2.3 Constraints on output of open-loop pumped storage.

Open-loop PHS is modeled as sharing a reservoir with hydropower to enhance the storage capacity of the hydropower system. For simplicity in description, we formulate the open-loop PHS with the assumption that the reservoir of pumped storage is higher than that of hydropower. In the model implementation, if the reservoir of storage is lower, we reverse the up-down direction. The power output to ($p_{g,k,t}^{\text{ol,dis}}$) and used from ($p_{g,k,t}^{\text{ol,char}}$) grid are determined by the water flow ($q_{g,(i,k),t}^{\text{ol,s2p}}$ and $q_{g,(i,k),t}^{\text{ol,p2s}}$), respectively, as follows:

$$p_{g,k,t}^{\text{ol,dis}} = q_{g,(i,k),t}^{\text{ol,s2p}} \times H_{g,k}^{\text{ol}} \times \eta_{\text{phs}} \times \rho \times g_e, \quad \forall g, k, t, \quad (5)$$

$$p_{g,k,t}^{\text{ol,char}} = q_{g,(i,k),t}^{\text{ol,p2s}} \times H_{g,k}^{\text{ol}} \times \frac{1}{\eta_{\text{phs}}} \times \rho \times g_e, \quad (6)$$

$$0 \leq p_{g,k,t}^{\text{ol,dis}}, p_{g,k,t}^{\text{ol,char}} \leq z_{g,k}^{\text{ol}} \times \bar{p}_{g,k}^{\text{ol}}, \quad (7)$$

where $\langle i, k \rangle$ denotes the pair of i -th reservoir hydropower and k -th open-loop PHS, $q_{g,(i,k),t}^{\text{ol,s2p}}$ ($q_{g,(i,k),t}^{\text{ol,p2s}}$) is water flow ($\text{m}^3 \text{s}^{-1}$) from storage (hydropower) to the hydropower (storage), $H_{g,k}$ is the elevation difference (m) between storage and hydropower, η_{phs} is the output efficiency of pumped storage, $z_{g,k}^{\text{ol}} \in [0,1]$ and $\bar{p}_{g,k}^{\text{ol}}$ are the installation factor and installation potential capacity of open-loop PHS k in planning year, respectively.

4.2.4 Constraints on cascade reservoir operation of hydropower and open-loop pumped storage. The reservoir capacity $v_{g,i,t}^{\text{hp}}$ of hydropower i in province g at time t is constrained by the capacity of previous time $v_{g,i,t-1}^{\text{hp}}$ and net-in water volume as follows:

$$v_{g,i,t}^{\text{hp}} = v_{g,i,t-1}^{\text{hp}} + [q_{g,i,t}^{\text{hp,in}} - q_{g,i,t}^{\text{hp,gen}} - q_{g,i,t}^{\text{hp,spill}} + q_{g,(i,k),t}^{\text{ol,s2p}} - q_{g,(i,k),t}^{\text{ol,p2s}}] \times \Delta_t, \quad \forall g, i, t, \quad (8)$$

where Δ_t is 3600 seconds within an hour, $q_{g,i,t}^{\text{hp,in}}$ is water inflow, $q_{g,i,t}^{\text{hp,spill}}$ is the released flow over spillways, and $q_{g,(i,k),t}^{\text{ol,s2p}} - q_{g,(i,k),t}^{\text{ol,p2s}}$ is net-in water flow from the paired open-loop PHS if equipped. The water inflow to hydropower i at time t contains two parts: one is the natural water inflow from upstream rivers, and the other is the outflow from upstream hydropower plants according to the cascade topology:

$$q_{g,i}^{\text{hp,in}} = \sum_{l \in L_{g,i}} q_{l,t-\tau_l}^{\text{nat}} + \sum_{j \in U_{g,i}} (q_{g,j,t-\tau_j}^{\text{hp,gen}} + q_{g,j,t-\tau_j}^{\text{hp,spill}}), \quad \forall g, i, t, \quad (9)$$

where τ is the delay time (h) from upstream rivers or hydropower plants, $\sum_{l \in L_{g,i}} q_{l,t-\tau_l}^{\text{nat}}$ is the sum of natural water inflow to hydropower i at time t from upstream rivers, $L_{g,i}$ is the set of upstream rivers, and $U_{g,i}$ is the set of upstream hydropower plants. The delay time τ is empirically derived *via* cross-correlation analysis⁵¹ on 3-hourly historical discharge time series of the plant with each of its neighboring upstream sources (see SI Section 2.3 for details of cascade topology construction and delay time calculation). In this study, we

use the peak inflow (97th percentile of historical discharge)¹⁷ to constrain the water inflow for power generation. The spillage flow is constrained by the maximum capacity of slipways ($\bar{q}_{g,i,t}^{\text{hp,spill}}$) and environmental flow ($q_{g,i}^{\text{hp,env}}$) requirement:

$$q_{g,i}^{\text{hp,env}} \leq q_{g,i,t}^{\text{hp,spill}} \leq \bar{q}_{g,i}^{\text{hp,spill}}, \quad \forall g, i, t, \quad (10)$$

where the environmental flow requirement is the 30th percentile of historical discharge.¹⁷ Similar to the reservoir capacity operations of hydropower, the volume of water in open-loop PHS ($v_{g,k,t}^{\text{ol}}$) at time t is equal to the water stored at the previous time ($t - 1$) plus net-in water:

$$v_{g,k,t}^{\text{ol}} = v_{g,k,t-1}^{\text{ol}} + [q_{g,k,t}^{\text{p2s}} - q_{g,i,t}^{\text{s2p}}] \times \Delta t, \quad \forall g, k, t. \quad (11)$$

Water volume of reservoir for hydropower and pumped storage at each time t is required to be the same at the beginning ($t = 0$) and end ($t = T$) of the planning period and lie within the minimum reservoir level requirement ($V_{g,i}^{\text{ol}}$) and maximum reservoir capacity ($\bar{V}_{g,i}^{\text{ol}}$). These capacity limits are determined based on official government data, which implicitly incorporate long-term safety and flood control provisions (see SI Section 2.1 for details):

$$v_{g,i,0}^{\text{ol}} = v_{g,i,T}^{\text{ol}}, \quad \forall g, i, \theta \in \{\text{hp}, \text{ol}\}, \quad (12)$$

$$V_{g,i}^{\text{ol}} \leq v_{g,i,t}^{\text{ol}} \leq \bar{V}_{g,i}^{\text{ol}}, \quad \forall g, i, t, \theta \in \{\text{hp}, \text{ol}\}. \quad (13)$$

4.2.5 Constraints on closed-loop pumped storage. Closed-loop pumped hydro storage is modeled as an independent energy storage system consisting of two reservoirs with an elevation difference (upper and lower). The power output from the storage system to the grid (discharging) or from the grid to the storage system (charging) at time t is directly related to the water flow between the upper and lower reservoirs, denoted as $q_{g,s,t}^{\text{cl,u2l}}$ (flow from upper to lower) and $q_{g,s,t}^{\text{cl,l2u}}$ (flow from lower to upper), respectively. These relationships are expressed as follows:

$$p_{g,s,t}^{\text{cl,disc}} = q_{g,s,t}^{\text{cl,u2l}} \times H_{g,s}^{\text{cl}} \times \eta_{\text{phs}} \times \rho \times g_{\text{e}}, \quad \forall g, s, t, \quad (14)$$

$$p_{g,s,t}^{\text{cl,char}} = q_{g,s,t}^{\text{cl,l2u}} \times H_{g,s}^{\text{cl}} \times \frac{1}{\eta_{\text{phs}}} \times \rho \times g_{\text{e}}, \quad (15)$$

$$p_{g,s,t}^{\text{cl,disc}}, p_{g,s,t}^{\text{cl,char}} \leq z_{g,s}^{\text{cl}} \times \bar{p}_{g,s}^{\text{cl}}, \quad (16)$$

where $p_{g,s,t}^{\text{cl,disc}}$ ($p_{g,s,t}^{\text{cl,char}}$) is the power from closed-PHS (grid) to grid (PHS), $H_{g,s}^{\text{cl}}$ is the elevational difference between upper and lower reservoirs, and $z_{g,s}^{\text{cl}} \in [0,1]$ and $\bar{p}_{g,s}^{\text{cl}}$ are the installation factor and installation capacity potential of the s -th closed-loop PHS in grid g , respectively. The water volume in the upper ($v_{g,s,t}^{\text{cl,up}}$) and lower ($v_{g,s,t}^{\text{cl,lo}}$) reservoirs at time t is determined by the initial water volume plus the net inflow from the opposite reservoir. Similar to conventional hydropower reservoir control constraints, the dynamics of the water volumes are constrained by the following equations:

$$v_{g,s,t}^{\text{cl,up}} = v_{g,s,t-1}^{\text{cl,up}} + (q_{g,s,t}^{\text{cl,l2u}} - q_{g,s,t}^{\text{cl,u2l}}) \times \Delta t, \quad \forall g, s, t, \quad (17)$$

$$v_{g,s,t}^{\text{cl,lo}} = v_{g,s,t-1}^{\text{cl,lo}} + (q_{g,s,t}^{\text{cl,u2l}} - q_{g,s,t}^{\text{cl,l2u}}) \times \Delta t, \quad \forall g, s, t, \quad (18)$$

$$v_{g,s,0}^{\text{ol}} = v_{g,s,T}^{\text{ol}}, \quad \forall g, s, \theta \in \{\text{up}, \text{lo}\}, \quad (19)$$

$$V_{g,s}^{\text{ol}} \leq v_{g,s,t}^{\text{ol}} \leq \bar{V}_{g,s}^{\text{ol}}, \quad \forall g, s, t, \theta \in \{\text{up}, \text{lo}\}. \quad (20)$$

Besides electricity supply, closed-loop PHS can also provide inertia and spinning reserve capacity for system security requirements.

Enhanced by the detailed modeling of hydropower and pumped storage systems—both open- and closed-loop—at the installation level, coupled with the co-optimization of capacity expansion and hourly cascade operations across an entire planning year, the scale of the CISPO developed in this study demands roughly 1.2 TB of memory and nearly 30 days of processing time to solve.

4.3 Electricity demand projection

In this study, the hourly electricity demand is projected to 2060 by the product of the national demand increase ratio and the provincial demand profile in 2019.⁴ To address the complex socioeconomic factors inherent in electricity demand forecasting, we develop our 2060 national electricity demand projections through a comprehensive methodological approach that integrates: (1) predictions from regression models constructed using historical consumption patterns; (2) simulated outcomes from economy-wide and integrated assessment models; and (3) insights derived from an extensive literature review and expert elicitation processes. For the provincial-level regression model, we use the historical population and gross production by sector as independent variables, and the yearly electricity as the dependent variable (2000–2020) for model fitting (most R^2 are larger than 0.95), and use forecasted independent variables^{52,53} under different SSP scenarios^{54,55} to project the national electricity demand in 2060. Across SSP1–SSP5 scenarios, we find that the projected total electricity demand in this method ranges from about 17.2 to 27.4 PWh per year. Economy-wide models report that China's electricity in 2060 will be 15.0–20.0 PWh per year.³⁴ Simulation results from two IAMs (REMIND1.7 and WITCH2016) in the ADVANCE database⁵⁶ predict the values of electricity consumption in 2060 from 16.7 to 21.5 PWh per year. Literature reviews and expert judgment project that the yearly electricity demand is from 15.7 to 22.2 PWh per year.^{57–59} Combining these findings (15.0–27.4 PWh per year), we adopt 20.0 PWh per year as the base electricity demand assumption in 2060.

Author contributions

Z. Z.: conceptualization, methodology, investigation, software, data collection and calibration, formal analysis, visualization, writing – original draft preparation, writing – review and editing; H. M.: data collection and calibration, software, visualization, writing – review and editing; S. Z., X. H.: writing – review and editing; D. Z.: conceptualization, writing – review and editing, funding acquisition, supervision.

Conflicts of interest

There are no conflicts to declare.

Data availability

All data used in this study are reported in the paper or supplementary information (SI). Supplementary information is available. See DOI: <https://doi.org/10.1039/d5ee05948h>.

The installed and potential hydropower resources generated in this study could be found at GitHub⁶⁰ (<https://github.com/mrziheng/HydroCHN>).

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